LEE MARQUEZ

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EDUCATION

Massachusetts Institute of Technology — Cambridge, MA
Bachelor of Science in Aeronautical and Astronautical Engineering — *GPA: 5.0/5.0* — August 2025 – May 2029

- **Relevant Coursework:** Differential Equations, Materials and Structures, Signals and Systems, Math for CS
- **Activities/Societies:** Students for the Exploration and Development of Space, Society of Hispanic Professional Engineers

EXPERIENCE

MIT REALM Lab (Prof. Chuchu Fan) — Cambridge, MA
Undergraduate Researcher – UROP — Fall 2026 – Present

- Researching vision-language model (VLM) and vision-language-action (VLA) planning for robot navigation.
- Developing short-horizon and long-horizon VLM planning for a ground vehicle/quadruped to reach end goals from images and video of its environment.
- Specifying tasks formally with Signal Temporal Logic (STL), such as ordered visits to locations followed by a task.

Universidad Panamericana — Mexico City, MX
Research Intern – Hybrid Rocket Propulsion — Summer 2026

- Designed and optimized a hybrid rocket engine using paraffin wax fuel and nitrous oxide (N_2O) oxidizer.
- Simulated performance in the Hybrid Rocket Analysis Program (HRAP), tuning injector diameter to improve oxidizer utilization, peak thrust, and total impulse.
- Worked within fixed design constraints (3.5 in maximum grain outer diameter, 10 lbm oxidizer charge).

MIT Design/Build/Fly — Cambridge, MA
Aerodynamics Subteam Member — Fall 2026 – Present

- Work on the aerodynamics of the competition aircraft, including the airframe, wings, and other aerodynamic surfaces.
- Contribute to the design of the aircraft as part of the aerodynamics team within a multidisciplinary build team.

MIT Rocket Team — Cambridge, MA
Propulsion Subteam Member — Fall 2025

- Design and test components tester to characterize throttle valve for self-landing rocket project.
- Draw Piping and Instrumentation Diagram (P&ID) for components tester.
- Present design in design reviews to industry.

PT Industries Inc. (Aerospace Contractor) — Signal Hill, CA
Intern – Quality Control Department — Summer 2023

- Inspected aerospace parts using micrometers, digital calipers, and other measurement tools.
- Programmed Coordinate Measuring Machine (CMM) using PC-DIMIS software to inspect parts.
- Operated CNC machines (waterjet, CNC mill) to manufacture aerospace parts.

Projects

NASA Student Launch Initiative — Long Beach, CA
Co-Captain (2024–2025), Safety Officer (2021–2023) — September 2021 – June 2025

- Led team of 8 to design, build, and launch a high-power rocket; managed 5 subteams, project timelines, and updates to the NASA Student Launch Board.
- Conducted research with a biological payload experiment, winning first place for NASA Student Launch Payload Design.
- Presented to NASA employees in design review meetings.
- Used OpenRocket to simulate rockets before launches; created safety matrices, checklists, and rules.

American Rocketry Challenge — Long Beach, CA
Co-Captain (2023–2025), Member (2018–2022) — August 2018 – June 2025

- Competed against 800 other teams in a model rocketry competition; placed top 25 at the National Finals in Washington D.C. 4 times and top 100 out of 800 for six years.
- Designed rockets to competition requirements utilizing 3D printing and laser cutting machines.
- Placed 8/800 in the American Rocketry Challenge.

SKILLS AND CERTIFICATIONS

Software: Fusion 360, Inventor, SolidWorks, OpenRocket, HRAP, PC-DIMIS, Python
Fabrication: 3D printing, laser cutting, CNC milling, waterjet, CMM
Certifications: OSHA General Industry Safety, NAR Level 1
Skills: CAD/CAM, Piping and Instrumentation Diagrams (P&ID)